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SYSTEM DEVELOPMENT LIFE CYCLE

System Development Life Cycle

- ✓ The process of developing new information systems is often called the system development life cycle.
- ✓ It basically includes a systematic process of planning, designing, and creating an information system for organizations.
- ✓ SDLC uses a systems approach for problem solving that basically states that complex problems need to be broken up into **smaller manageable problems** using a **systems' hierarchy**, and then developing a solution for each problem within the hierarchy.

- ✓ The SDLC consists of tasks that are **divided into phases or stages** as shown in Figure 4-1. Please read systems analysis and design books for complete details on SDLC.

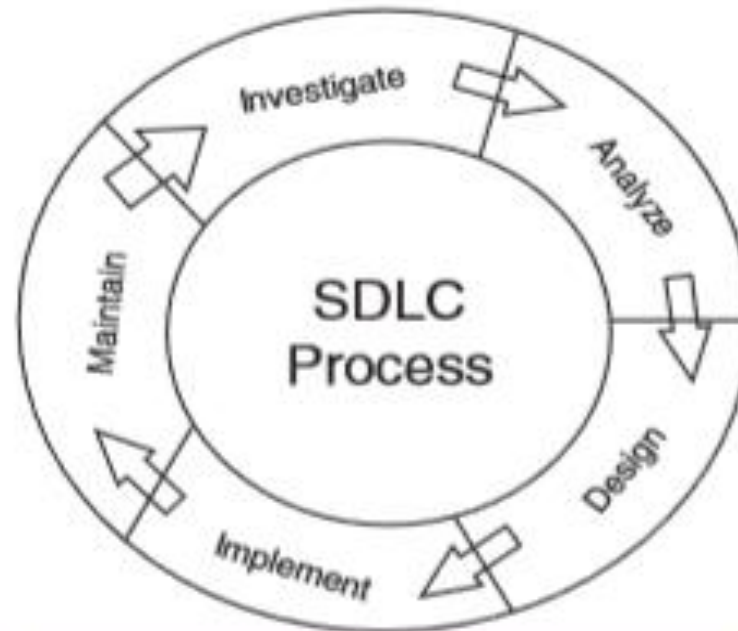


FIGURE 4-1 Traditional SDLC Methodology

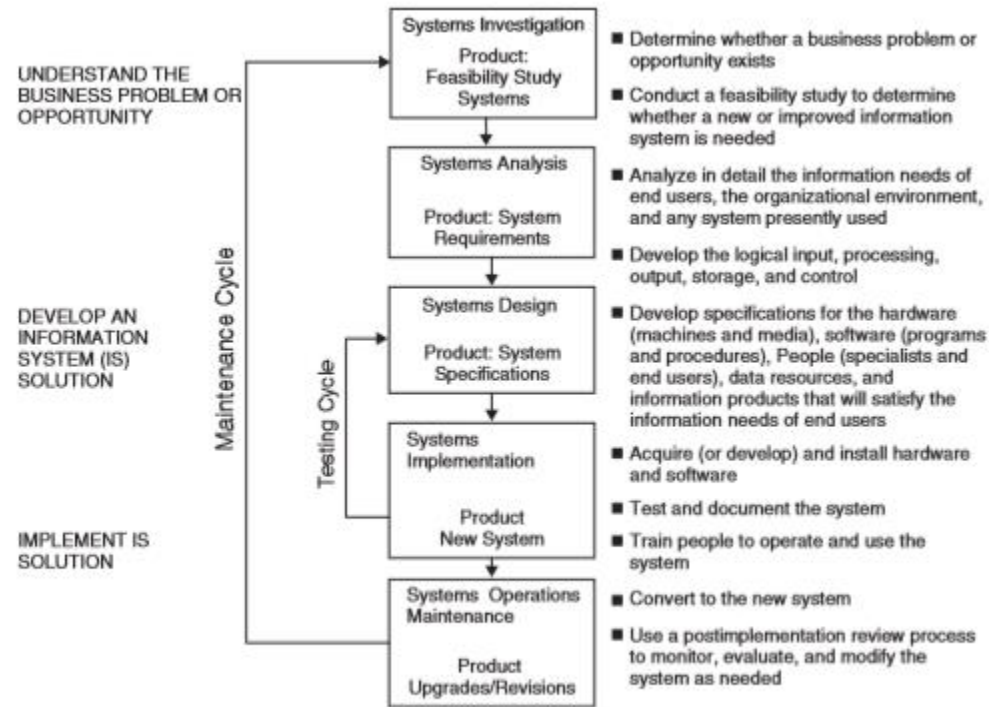


FIGURE 4-2 SDLC Approach

Figure 4-2 provides a summary of the traditional SDLC methodology. The SDLC process begins when someone in the organization identifies a need, or investigation phase, for a new system and ends the implementation phase where the new system is installed and running in the organization.

RAPID SDLC APPROACHES

- ✓ **Rapid development** approach is *prototyping* This approach does not go through the analysis and design phases; instead, it implements a skeleton or a prototype of the actual system with a focus on input (i.e., user interface) and output (i.e., screen displays and reports generated with dummy data).

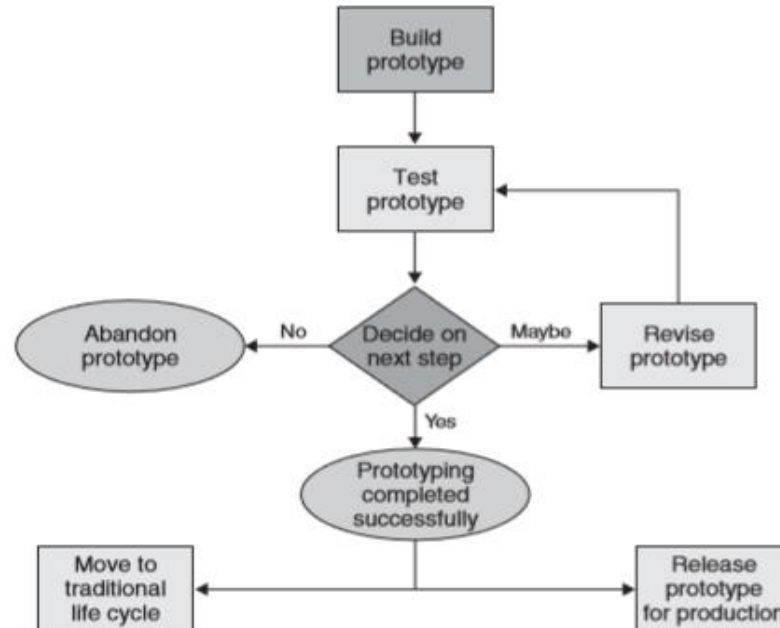


FIGURE 4-3 Prototype Development



✓ Another rapid development approach is ***end-user development (EUD)***, which lets the end users **create their own applications**. This process became popular in the **1980s** with the advent of personal computers (PCs). In this process the users are trained by the IT staff or professional trainers to develop customized applications (e.g., a small decision-making application with an Excel spreadsheet or a departmental employee tracking system with an Access database).

ERP IMPLEMENTATION LIFE CYCLE

- ✓ ERP applications are prepackaged software developed by commercial software vendors and custom installed for organizations to automate and integrate the various business processes.

TABLE 4-1 Differences Between ERP and Other Software Packages

	ERP Software	Other Packaged Software
Software Cost	Millions of dollars	Hundreds to thousands
Significance to Organization	Mission critical	Support or productivity improvement
Installation Time	One to several years	Almost instantly
Change Management Strategy	Requires significant change management strategy from beginning to end for success; business process change, training, communications, etc.	Requires some training and support
Implementation Costs	Requires in-house employee time, consultants, and vendor support in millions of dollars	Requires little or no consulting support or vendor technical support

These are complex software packages **costing millions of dollars** to develop and maintain that automate hundreds of business processes in an organization.

ERP IMPLEMENTATION PLAN

- ✓ An ERP implementation plan is used to create a **roadmap or blueprint** to meet **cost, scope, and time constraints of an implementation**. There are many different ERP implementation methodologies promoted by different vendors and consultants. The appropriateness of the plan depends, in part, on the project, the company, and the reasons for the implementation.

Following are three major implementation plan choices:

- 1. Comprehensive-** A comprehensive ERP integration plan is the most expensive, lengthy, and costly approach. It involves implementation of the full functionality of the ERP software in addition to industry-specific modules. Implementing the full functionality requires a high level of business process reengineering (BPR) with major changes in the business processes and customization of legacy systems.

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2. **Middle of the Road-** A middle-of-the-road ERP implementation plan involves some changes in the core ERP modules and a significant amount of BPR. The middle-of-the-road approach is not as expensive as the comprehensive approach or as straightforward as the vanilla approach.
 3. **Vanilla-** A vanilla ERP implementation plan utilizes core ERP functionality and exploits the best practice business processes built into the software. A company following a vanilla implementation will have to simply align their business processes to the ERP system, rather than modify the software. By eliminating or minimizing the required BPR, the project's costs and time required for the implementation are minimized

ERP IMPLEMENTATION METHODOLOGY

- ✓ Methodology refers to a ***systematic approach*** to solving a business problem. ERP methodology builds on the theory that an enterprise can maximize its returns by maximizing the utilization of its fixed supply of resources.
- ✓ The traditional ERP life cycle accomplishes one stage at a time and requires formal milestone approvals prior to moving to the next stage.

They also allow flexibility and quicker feedback loops to accommodate rapid corrections as shown in Figure 4-4.

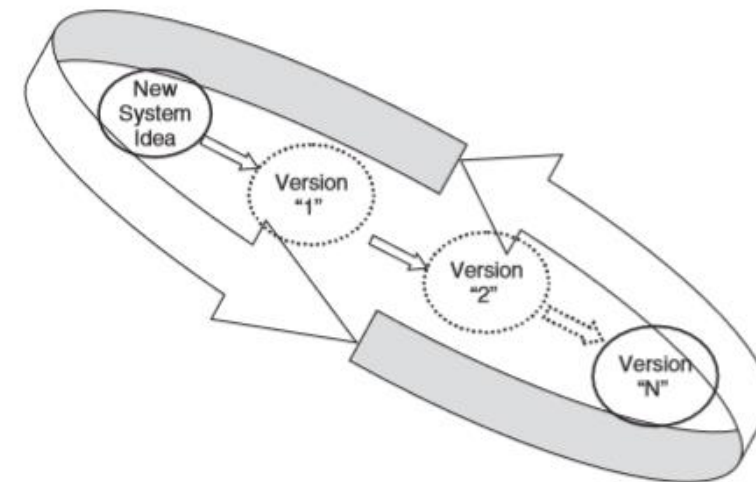


FIGURE 4-4 Rapid Application Development Process



TRADITIONAL ERP LIFE CYCLE

- ✓ The traditional ERP life cycle approach has a **deliverable at the end of each stage** (e.g., a report with supporting documents) that is reviewed by management and upon which a decision is made either to continue with the project or not.
- ✓ The emphasis in ERP implementation is **on customizing the software** as well as on **changing the organization's business processes**, rather than determining the user requirements for developing new applications (as in the traditional SDLC).

- ✓ ERP life cycle, as shown in **Figure 4-5**, iterates a

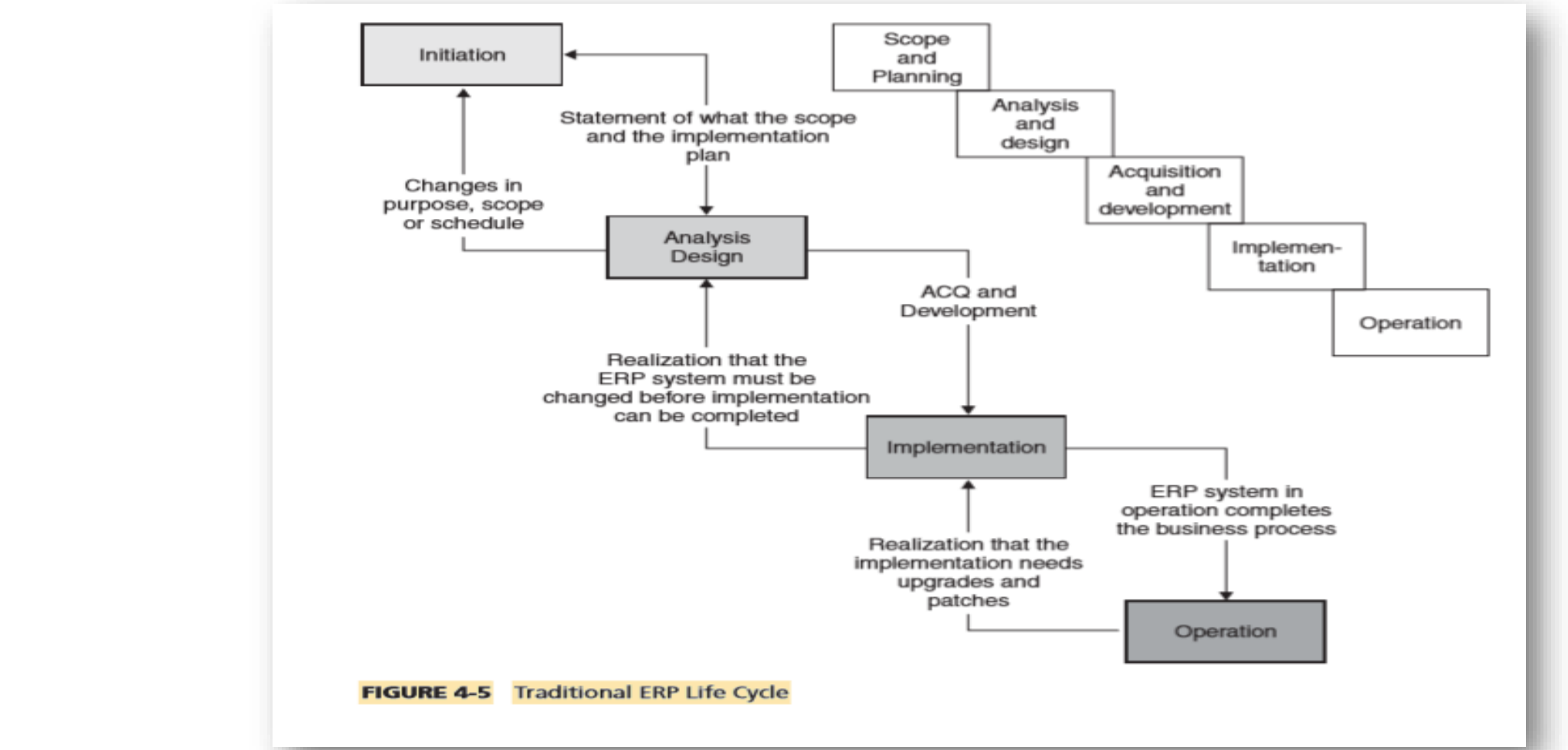


FIGURE 4-5 Traditional ERP Life Cycle

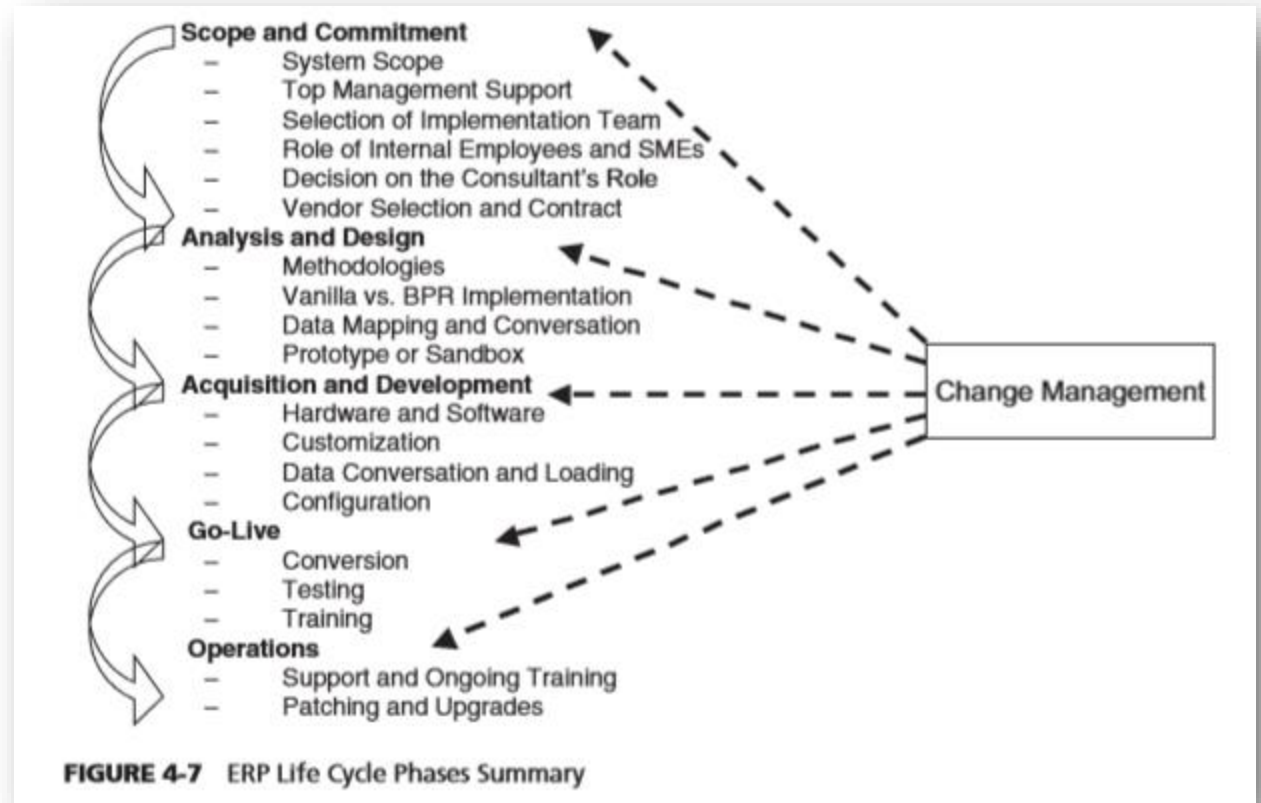
The **traditional ERP life cycle** includes the following major stages:

Stage 1. Scope and commitment stage

Stage 2. Analysis and design stage

Stage 3. Acquisition and development stage

Stage 4. Implementation stage



ERP LIFE CYCLE VS. SDLC

Some of the key differences are as follows:

- ✓ SDLC does not mention software acquisition until the fourth stage, whereas in the ERP life cycle, the ERP software must be selected at a very early stage of the implementation process. One of the key early decisions in the ERP life cycle is software or vendor selection. ERP vendors have traditionally embedded the best practices and business rules in their software. Some vendors specialize in certain industries. Understanding the ERP software's functionality and the embedded business processes are therefore crucial for successful implementation. A good match between the company's business process and software's embedded functionality means quicker implementation and millions of dollars saved in implementation costs.
- ✓ In SDLC the new application is custom designed based on the user requirements as determined from the feasibility study and analysis. On the other hand, in the ERP life cycle the new application is bought by the organization and users are asked to change their business process and policy to take advantage of the best practices embedded in the ERP software. The emphasis in ERP life cycle is more toward reengineering organizational process and change management to improve productivity and create efficiencies with the help of embedded functionality of the ERP software.
- ✓ Another difference is in the role of external consultants in the ERP life cycle. In traditional SDLC, the consultant's role is limited to IT hardware, software, and training. Most of the team is made up of people from inside the organization. Consultants play a very important role right from beginning to end during ERP installations advising the organization on software vendor selection, business process reengineering, software installation, and change management.

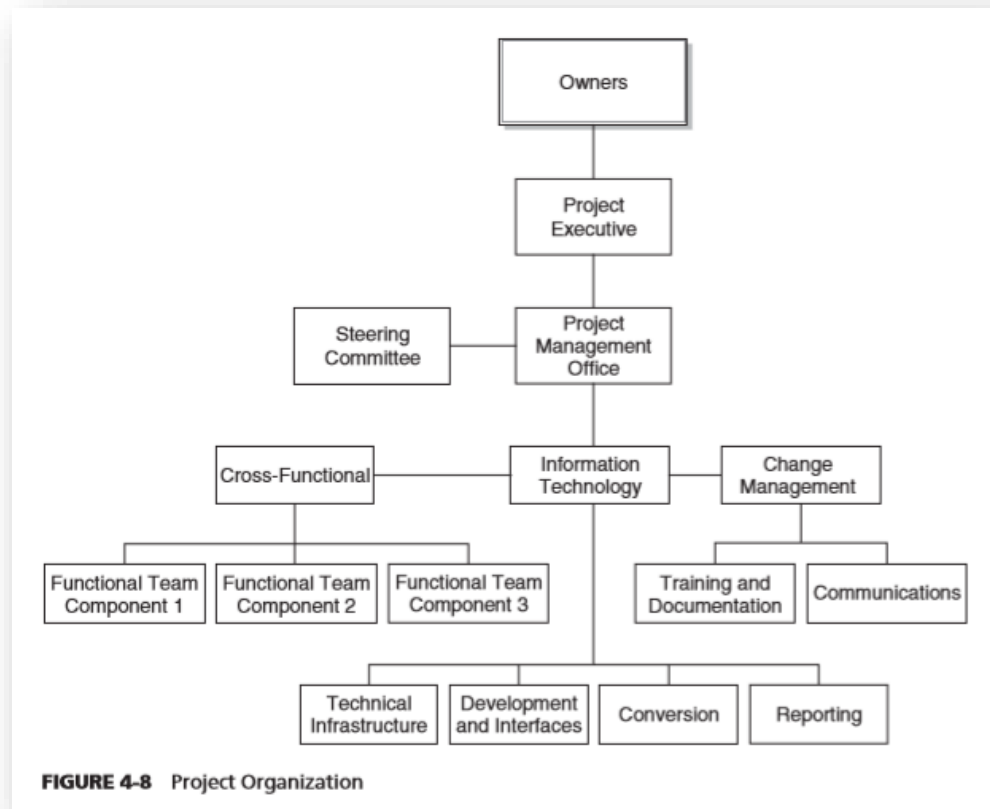
TABLE 4-3 Comparing and Contrasting SDLC with ERPLC

	SDLC	ERP Life Cycle
Goal	Develop a new system to support the organization requirements	Implement a packaged system to support the organization requirements
Analysis	Evaluate user needs through observations and interviews and create system specifications	Vendor analysis and evaluation of business process changes due to the implementation
Design	Develop new system architecture, user interface, and reporting tools	Installation and customization plan of ERP software, data conversion, and change management strategies
Implementation	Acquire hardware, software, develop applications, installation, testing, training, and conversion	"Go-Live" conversion or releasing the system to the users, training, and support
Consultant Role	Technical support mainly during design and implementation	Change management, process change, and technical support from beginning to end
Management Role	Some oversight and support	Significant oversight and involvement—especially in change management
End-User Role	Focus group providing input during the various stages with most involvement during implementation stage	Multiple groups such as SMEs, advance users, and self-service users are part of implementation team with continuous involvement
Operations	Maintains, updates, and provides technical support	Maintains, updates, upgrades, and monitors change management strategy

There are also differences due to the prepackaged nature of the software as shown in Table 4-3.

PROJECT MANAGEMENT

- ✓ ERP projects take on their own organizational duties and job functions separate and apart from the day-to-day business functions. A clear project plan and reporting structure will better ensure that the project receives the attention and accountability needed to be successful.



The **organizational structure** must coincide with the **project governance**, **project owners**, **project steering committee**, and **project executive** must develop the hierarchy and determine **who is responsible** for each system implementation component.

PROJECT ROLES AND RESPONSIBILITY

TABLE 4-4 Summary of Roles for ERP Implementation

Project Role Title	Role Definition
Owners	The owners will consist of senior management in the company. The chair is empowered to make decisions when the owners cannot reach consensus. The owners determine overall policy, budget, and scope of the project. The owners meet when needed at the call of the chair.
Project Executive	The project executive oversees project activities, provides broad project oversight, resolves policy level issues, and ensures that the project stays within scope. The project executive also builds consensus on business process changes that impact the business and provides project status updates (as needed) to the owners. The project executive works with the application steward and project manager(s) to establish overall project direction, review and evaluate project progress, and ensure appropriate user involvement for the duration of the project.
Project Manager(s)	The project manager manages the day-to-day aspects of the project, ensures that the project plan is being followed, and keeps both team members and the project executive aware of the status of the project. This responsibility includes overall management of the project to ensure that all tasks are completed on a timely basis, in a quality fashion, and in accordance with the approved project plan. The project manager serves as the primary liaison between the project team and project executive.
Functional Module Leads	The module leads provide leadership and overall direction for the implementation at the module level, ensuring the quality of deliverables and adherence to the project plan and milestones.
Technical Infrastructure Team Lead	The technical infrastructure lead is responsible for overall technical infrastructure implementation including organizing and directing the efforts of the team, coordinating technical team meetings, and reporting on progress to project executive and project manager(s). The technical infrastructure lead must understand all of the technologies well enough to ensure that there will be no "gaps" in the solution and that all of the technologies will be integrated.
Development Lead	The development lead is responsible for managing the development team in the design and implementation of all modifications including reports, interfaces, online changes, and batch programs.
Change Management Lead	The change management lead coordinates the overall change management effort, including training, communication, and campus readiness activities. The change management lead is responsible for developing strategies and detailed work plans, monitoring progress, and resolving change management issues.
Conversion Lead	The conversion lead coordinates the overall conversion effort. The conversion lead is responsible for developing strategies and detailed work plans, monitoring progress, and resolving issues.

TABLE 4-4 Continued

Project Role Title	Role Definition
Reporting Lead	The reporting lead coordinates the overall reporting effort. The reporting lead is responsible for developing strategies and detailed work plans, monitoring progress, and resolving issues.
System Test Lead	The system test lead coordinates the overall system testing effort. The system test lead is responsible for developing strategies and detailed work plans, monitoring progress, and resolving issues.
Module Team Members	Module team members are responsible for analyzing requirements and converting them into solutions. Module team members provide direction and application knowledge with respect to business process design, configuration, testing, training, and implementation. Module team members are composed of the functional and technical resources from the project and campuses and report to the module leads.
Technical Infrastructure Team Members	The technical infrastructure team members are responsible for supporting the module teams throughout the design and implementation of the ERP software. The technical infrastructure team members are responsible for designing and building the necessary architecture components to supplement the delivered ERP technical solution.
Development Team Members	Development team members are responsible for developing the designed solutions necessary to meet business requirements. Development team members provide direction and ERP technical knowledge to the client team with development of modifications and interfaces.
Change Management Team Members	Change management team members are responsible for completing the tasks related to the training development and delivery.
Conversion Team Members	Conversion team members are responsible for designing and developing conversion programs (temporary and permanent) necessary to convert the legacy system data into the ERP database.
Reporting Team Members	Reporting team members are responsible for designing and developing the reports needed to support the system.
System Test Team Members	System test team members are responsible for completing the tasks related to system test effort.
Subject Matter Experts (SMEs)	Subject matter experts are responsible for ensuring that business-specific requirements are addressed in the design, build, and the system is tested. SMEs provide coordination and facilitation of communications between the project team and the organization. SMEs coordinate and prioritize functional requirements. SMEs provide leadership and functional expertise in support of the implementation with specific knowledge in one or more business processes.
Project Administrative Support	Project administrative support is responsible for providing administrative to the project.



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